Table 11.2 Classes of organic compounds based on functional group

Class of the compound	Functional group	Common Formula	Examples
Alcohol	–OH	R-OH	Ethanol, CH_3CH_2OH
Aldehyde	O ‖ –C–H	R-CHO	Acetaldehyde, CH_3CHO
Ketone	O ‖ –C–	R–CO-R	Acetone, CH_3COCH_3
Carboxylic acid	O ‖ –C–OH	R-COOH	Acetic acid, CH_3COOH
Ester	O ‖ –C–OR	R-COOR	Methyl acetate, CH_3COOCH_3
Ether	–O–R	R-O-R	Dimethyl ether, CH_3OCH_3

11.4 HOMOLOGOUS SERIES

Homologous series is a group or a class of organic compounds having same general formula and similar chemical properties in which the successive members differ by a -CH_2 group.

Let us consider members of alkanes given in Table 11.1. Their condensed structural formulas are given below:

Methane - CH_4
Ethane - CH_3CH_3
Propane - $CH_3CH_2CH_3$
Butane - $CH_3(CH_2)_2CH_3$
Pentane - $CH_3(CH_2)_3CH_3$

If you observe the above series. you can notice that each successive member has one methylene group more than the precedent member of the series and hence they are called homologs.

11.4.1 Characteristics of homologous series

- Each member of the series differs from the preceding or succeeding member by one methylene group ($–CH_2$) and hence by a molecular mass of 14 amu.
- All members of a homologous series contain the same elements and functional group.
- They are represented by a general molecular formula. e.g. Alkanes, C_nH_{2n+2}.
- The members in each homologous series show a regular gradation in their physical properties with respect to their increase in molecular mass.
- Chemical properties of the members of a homologous series are similar.
- All the members can be prepared by a common method.

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